

- 1 **NeuroPilot: Longitudinal intravital neural imaging across**
- 2 **scales and modalities via agentic platform**

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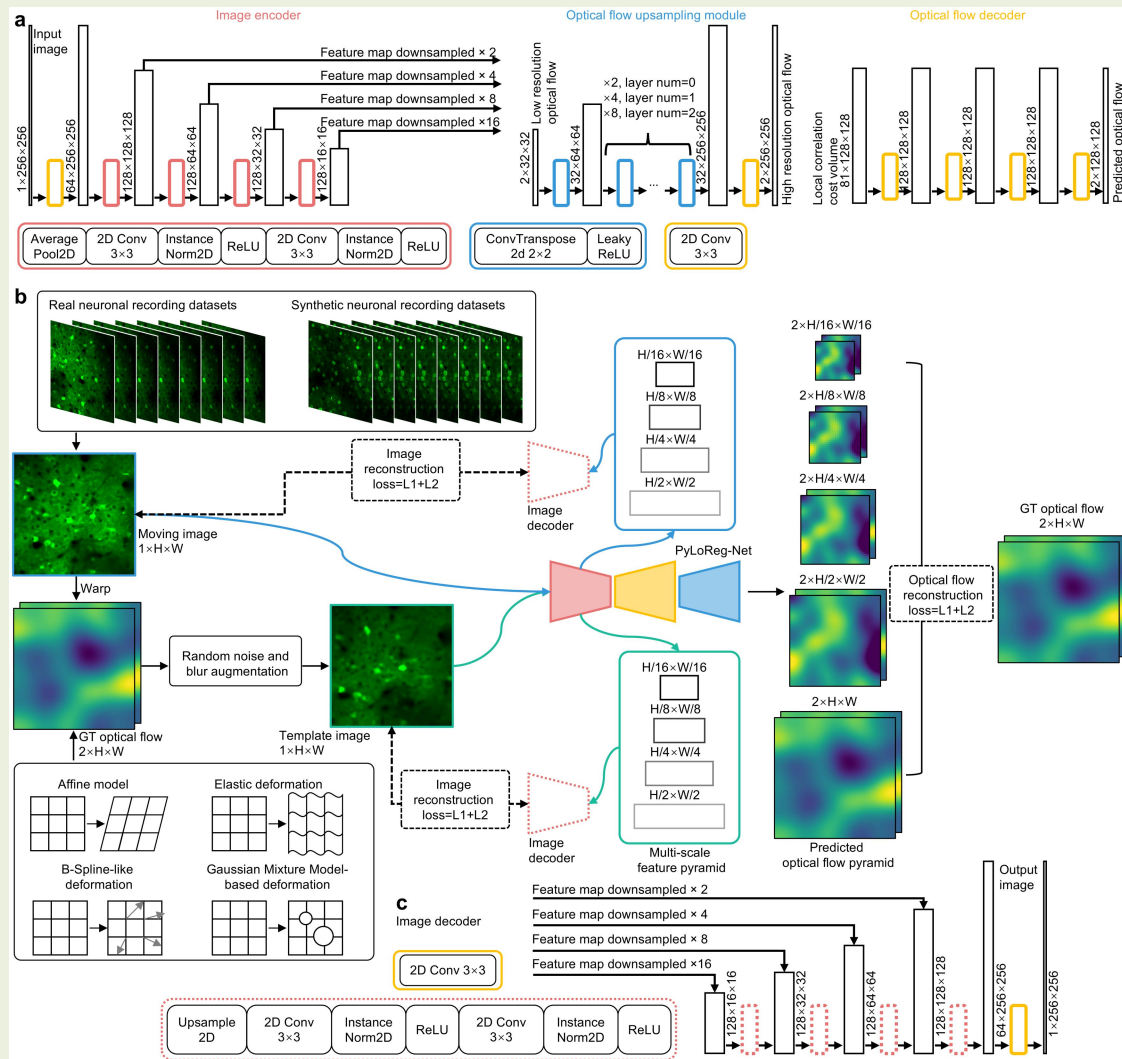
5 Supplementary Videos

Supplementary Video 1	Animated overview of NeuroPilot, illustrating LLM-orchestrated processing of neural imaging data through denoising, motion correction, segmentation, and analysis in a closed-loop pipeline. The video highlights data flow within PyLoReg-Net, where multi-scale features are iteratively refined to estimate dense non-rigid motion and progressively stabilize the imaging data.
Supplementary Video 2	4 stimulations This video demonstrates the stability of single-cell tracking in the mouse spinal cord across various external stimulations. Left: Illustrations showing physiological states, including free-moving, ice stimulations, mechanical stimulations, and spinal fixation. Middle and Right: Synchronized raw and NeuroPilot-processed recordings. Kymographs (bottom) demonstrate the superior temporal and spatial stability achieved after processing.
Supplementary Video 3	28 days This video demonstrates in vivo longitudinal imaging of spinal cord neurons in an SNI (spared nerve injury) model. Data was acquired via miniature two-photon microscopy over five recording sessions (Days 1, 5, 14, 21, and 28). Top row: Raw recordings showing progressive noise and signal degradation. Bottom row: NeuroPilot-processed outputs.
Supplementary Video 4	28 days seg This video demonstrates the longitudinal performance of NeuroPilot in segmenting individual spinal neurons and extracting calcium traces from free-moving mice across 28 days (Days 1, 5, 14, 21, and 28). Left: STD projection map overlaid with color-coded spatial masks for individual neuron

	identification. Middle and Right: Synchronized comparison of temporal calcium traces extracted from raw data (white) and NeuroPilot-processed data (green).
Supplementary Video 5	3P This volumetric video demonstrates the spatiotemporal restoration of low-signal-to-noise ratio (SNR) data acquired via miniature three-photon microscopy. Left: The raw 3D rendering exhibits significant noise and signal attenuation, particularly in deeper tissue layers. Right: The corresponding volume processed by NeuroPilot.
Supplementary Video 6	Large FOV NeuroPilot-enhanced large-scale cortical two-photon imaging. Real-time comparison between raw two-photon recordings and NeuroPilot-processed results. The video demonstrates signal restoration across a wide-field cortical area (encompassing a ~6 mm field of view) and within magnified local regions (regions 1-3).
Supplementary Video 7	Two color This video demonstrates the denoising and restoration capabilities of NeuroPilot on in vivo multiplexed two-photon microscopy data from mouse vibrissal primary somatosensory cortex (vS1, barrel cortex) neurons. The imaging captures both calcium dynamics (GCaMP6s, green) and structural morphology (mCherry, magenta TODO).
Supplementary Video 8	Vasculature High-fidelity restoration of spinal microcirculation in freely moving mice via NeuroPilot. This video demonstrates NeuroPilot's performance in restoring two-photon imaging data of the spinal cord vasculature from a freely moving mouse.

Supplementary Video 9	Deepbrain This video demonstrates stable, high-fidelity calcium imaging in two deep-brain regions of interest (ROIs) within mice, achieved through miniature two-photon microscopy and processed via NeuroPilot. Target Region 1: Lateral Entorhinal Cortex (LEC). Microendoscopy (via a GRIN lens) provided access to LEC neurons at a physical depth of 4.3 mm. Target Region 2: CA1 Regions. Imaging was performed at a physical depth of 1200 μm .
Supplementary Video 10	mitochondrial This video demonstrates the restoration of mitochondrial calcium dynamics and network topology from noisy two-photon microscopy data using NeuroPilot. Left: Fluorescence images comparing raw data (top) and NeuroPilot outputs (bottom). Insets highlight the recovery of continuous tubular structures. Right: Skeletonization analysis demonstrating that NeuroPilot accurately extracts the true mitochondrial network from noise.
Supplementary Video 11	24 h This video demonstrates continuous 24-hour calcium imaging of the medial prefrontal cortex (mPFC) in a freely behaving mouse, acquired via miniature two-photon microscopy at an imaging depth of 1800 μm . The video synchronizes a 24-hour circular ethogram with concurrent behavioral and neural data. Top panels display infrared behavior alongside Raw and NeuroPilot-processed calcium imaging. Extracted neuronal footprints and a normalized calcium heatmap are temporally aligned with the color-coded behavioral timeline.
Supplementary	Latent mPFC population dynamics during NREM-to-REM sleep transitions Continuous recording illustrating mPFC

Video 12	population activity and latent dynamics aligned to spontaneous NREM-to-REM transitions. Top: Behavioral annotation and circadian time (ZT). Middle: Representative behavioral frame, calcium imaging field, and segmented neuronal ROIs. Bottom left: Z-scored population activity aligned to the transition. Bottom right: Corresponding latent population trajectories inferred by rSLDS, overlaid on dynamical flow fields, revealing state-dependent reorganization of network dynamics.
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Supplementary Figure 1

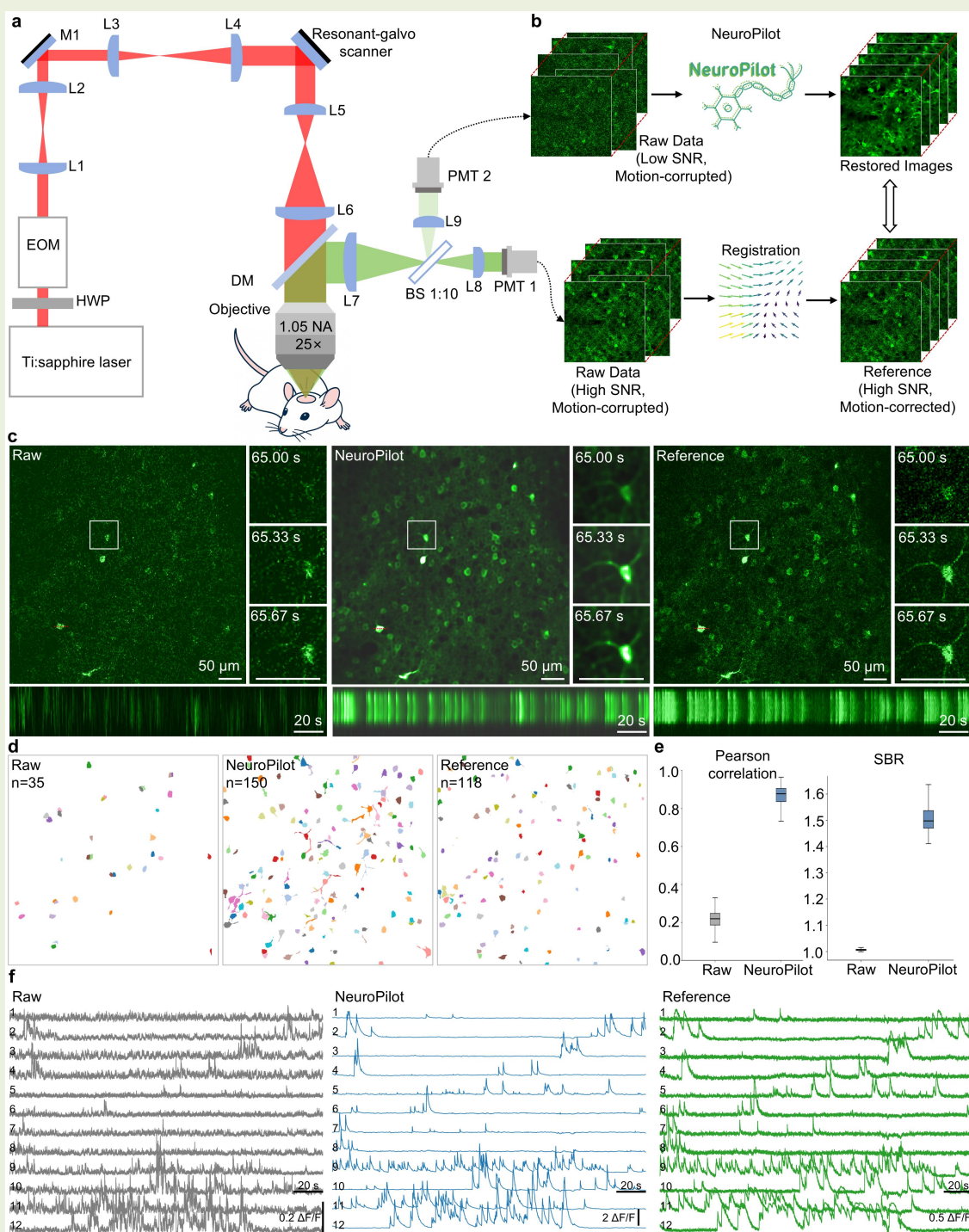
PyLoReg-Net architecture and training framework for learning-based non-rigid motion correction in NeuroPilot.

a, The image encoder of PyLoReg-Net extracts multi-scale feature pyramids from input images, producing feature maps downsampled by factors of 2, 4, 8, and 16 (intermediate channel width is 128). The optical flow decoder predicts flow fields from local correlation cost volumes via convolutional layers. The upsampling module refines low-resolution flow estimates using a sequence of transposed convolutions to generate progressively higher-resolution outputs.

b, PyLoReg-Net training framework. The training utilizes a hybrid dataset comprising both experimentally acquired and synthetically generated neuronal recordings. To

simulate realistic movement, a moving image frame is paired with complex ground truth (GT) optical flow synthesized through a combination of affine, elastic, B-Spline, and Gaussian Mixture Model-based (GMM-based) deformations. The corresponding template image is generated by warping the moving image with the GT flow, followed by random noise and blur augmentations. PyLoReg-Net processes this input pair to yield multi-scale feature pyramids and a predicted dense optical flow pyramid. Optimization is guided by minimizing a dual loss function: an image reconstruction loss ($L1+L2$), derived from reconstructing images from feature pyramids via the Image Decoder, and an optical flow reconstruction loss ($L1+L2$), calculated between the full-resolution predicted flow pyramid and the synthesized GT flow.

c, Architecture of the image decoder used during training. The decoder takes multi-scale feature maps from the encoder and progressively reconstructs a high-resolution image through a series of convolutional upsampling blocks. The reconstructed image is compared with the template image using an image reconstruction loss ($L1 + L2$), which encourages the encoder to learn informative pyramid features for accurate motion estimation.



Supplementary Figure 2

Two-photon imaging setup, paired datasets, and quantitative validation of NeuroPilot using in vivo recordings of neuronal somata.

a, Schematic of the two-photon microscopy system used for simultaneous acquisition of low- and high-SNR calcium imaging data. A Ti:sapphire laser, modulated by a half-

wave plate (HWP) and an electro-optic modulator (EOM), is delivered to the cortex through relay lenses (L1-L4) and a resonant-galvo scanner for high-speed two-dimensional imaging. Emitted fluorescence is collected through the objective and separated by a long-pass dichroic mirror (DM), then split by a beamsplitter (BS) into 10% and 90% channels. The two channels are detected by PMT1 and PMT2, yielding low- and high-SNR recordings, respectively. L1-L9, relay lenses; M1, mirror.

b, Workflow for generating paired validation datasets. Motion-corrupted high-SNR recordings were first corrected by NoRMCorre to generate a motion-corrected reference. In parallel, the corresponding low-SNR recordings were processed by NeuroPilot. The restored low-SNR data were then quantitatively compared with the high-SNR reference for performance evaluation.

c, Representative fields of view comparing raw recordings (left), NeuroPilot-restored outputs (middle), and high-SNR reference recordings (right) from *in vivo* imaging of neuronal somata. White boxes indicate regions shown at higher magnification in the adjacent insets at the indicated time points. Corresponding spatiotemporal kymographs are displayed below each field of view. Scale bars: 50 μm and 20 s.

d, Spatial footprints of neuronal somata segmented by Suite2p from raw recordings, NeuroPilot-restored outputs, and high-SNR reference recordings. The total numbers of detected neurons were 35, 150, and 118 for the raw, NeuroPilot, and reference data, respectively, indicating improved cell detectability after NeuroPilot restoration.

e, Quantitative evaluation of signal restoration. Boxplots show Pearson correlation coefficients between extracted calcium traces and the reference signals (left), and signal-to-background ratios (SBRs) for the raw and NeuroPilot-restored data (right).

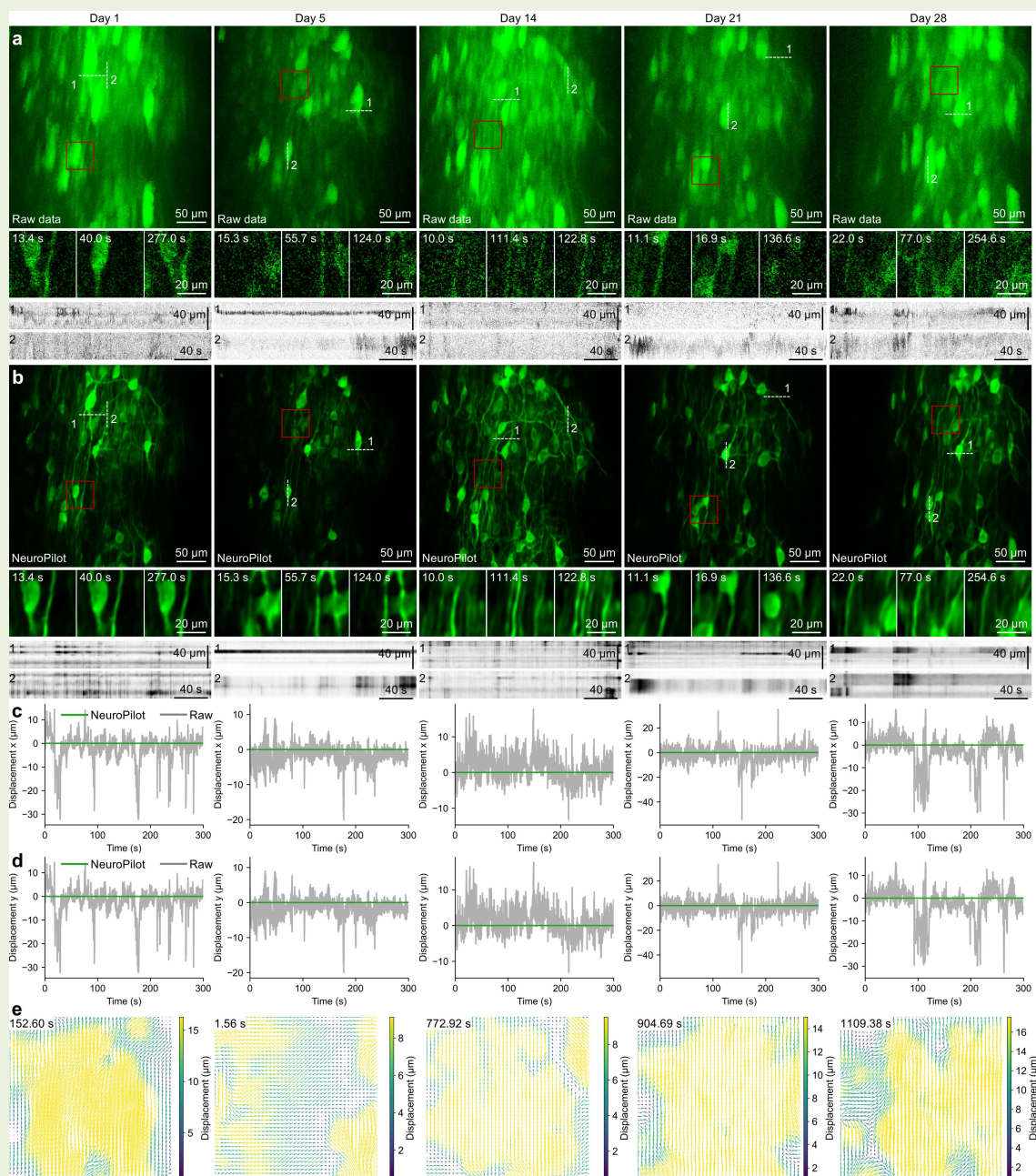
f, Representative calcium dynamics ($\Delta F/F$) from 12 selected neurons. Traces from raw recordings, NeuroPilot-restored outputs, and high-SNR reference recordings are shown from left to right. Scale bars: 20 s and 2 $\Delta F/F$.

Dataset name	Modality	Number of stacks	Number of frames	Image size/pixel	Usage in training	Notes
Synthetic dataset	Two-photon	200	200,000	490×490	All for training	90.5 GB
Neuronfinder	Two-photon	28	103,517	512×512	All for training	100.7 GB
CRCNS pvc-13	Two-photon	44	97,511	256×512	All for training	22.7 GB
CRCNS pvc-10	Two-photon	116	~541,000	256×455	Part for training	234 GB
CRCNS ssc-8	Two-photon	53	107,417	128×256	All for training	6.9 GB
CRCNS ssc-1	Two-photon	125	~350,000	512×512	Part for training	341 GB
CRCNS ssc-2	Two-photon	258	~558,000	512×512	Part for training	543 GB
CRCNS ssc-6	Two-photon	10	166,600	128×512	All for training	10 GB

Supplementary table 1

Summary of real and synthetic datasets used for training PyLoReg-Net.

Overview of the two-photon microscopy datasets utilized for network training. The training corpus comprises a large-scale synthetic dataset generated via NAOMI, alongside multiple empirical datasets from public repositories (Neuronfinder and CRCNS). The table outlines the specific characteristics of each dataset, including image size, total frame count, and overall data volume employed during the training phase.



Supplementary Figure 3

Cross-day stabilization of chronic spinal cord imaging across L3-L5 segments in freely moving mice.

a-b, Maximum intensity projections (MIPs) of longitudinal calcium imaging in a spared nerve injury (SNI) model across five sessions (Days 1, 5, 14, 21, and 28; Day 1, pre-injury baseline; Days 5 – 28, post-SNI). Raw recordings **a** are shown alongside NeuroPilot-restored outputs **b**. Bottom panels display magnified time-lapse frames from

the red-boxed regions and spatiotemporal kymographs extracted along the dashed lines (1 and 2). Scale bars: 50 μm (MIPs), 20 μm (insets), and 40 μm / 40 s (kymographs).

c-d, Quantification of image registration before and after motion-correction. Estimated frame-by-frame displacements along the x-axis **c** and y-axis **d** reveal severe motion artifacts in the raw data (gray traces), which are effectively suppressed to near-zero baselines by NeuroPilot (green traces).

e, Representative optical flow fields from individual sessions, illustrating the complex non-rigid motion patterns in chronic spinal cord imaging. Arrows indicate displacement direction, and color denotes displacement magnitude.

NeuroPilot performance across longitudinal one-photon imaging, three-photon volumetric imaging, and large-field-of-view two-photon imaging.

a, Representative field of view (FOV) comparing raw recordings (left) and NeuroPilot-restored results (right). Bottom insets show magnified time-lapse frames of the red-boxed regions at selected time points. Scale bars: 200 μm (main), 50 μm (insets).

b, Spatial footprints of segmented neuronal regions of interest (ROIs). Each color denotes an individual neuron.

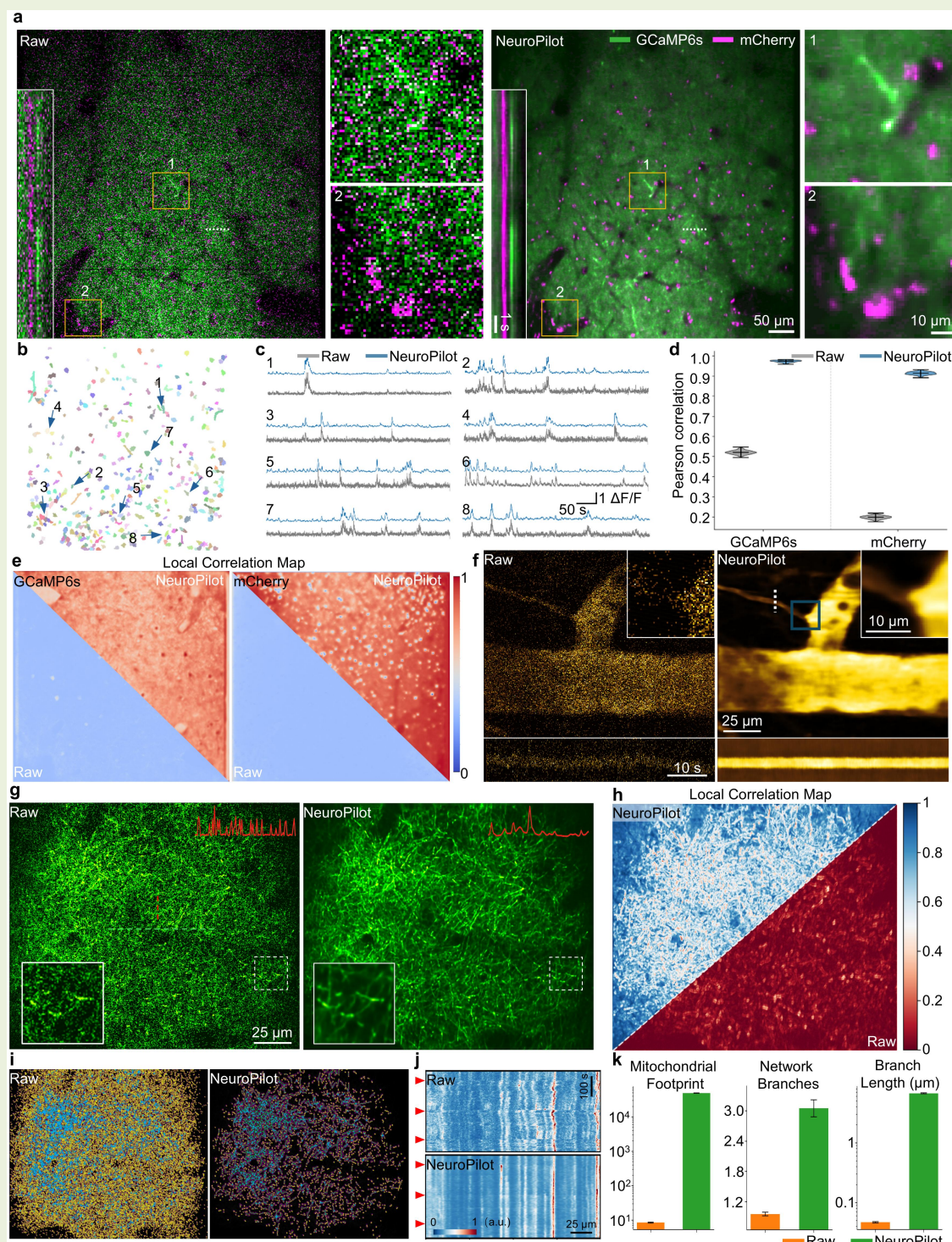
c, Example fluorescence traces ($\Delta F/F$) from selected neurons. NeuroPilot-processed signals (green) exhibit reduced baseline fluctuations and improved signal-to-noise ratio while preserving calcium transient amplitudes and temporal dynamics (arrowheads).

d, Kymographs extracted along dashed lines 1 and 2 in **a**, showing spatiotemporal signal propagation. Raw data display substantial background fluctuations and motion-induced distortions (arrowheads), which are effectively suppressed after NeuroPilot processing, resulting in enhanced temporal coherence. Scale bars: 100 μm ; 20 s.

e-h, Matched recordings from the exact same FOV acquired on Day 3, presented in the identical format to **a-d**.

i, Head-mounted three-photon volumetric imaging spanning cortical and hippocampal depths (~ 1000 μm axial range). Left, Single-volume low-SNR raw data acquired from GCaMP6s-expressing neurons in the posterior parietal cortex (PPC) and hippocampal CA1. Middle, High-SNR reference obtained by averaging 50 consecutive volumes, providing an estimate of the underlying ground-truth structure. Right, NeuroPilot-restored volume reconstructed from a single low-SNR stack, showing substantial recovery of neuronal signals and suppression of background noise. Cortical and subcortical layers (L2/3, L4/5, L6, corpus callosum (CC), and CA1) are annotated, demonstrating accurate reconstruction of depth-resolved neuronal structures across the full axial extent, with fidelity comparable to the averaged high-SNR reference.

j, Large field-of-view two-photon calcium imaging of the mouse cortex. Left, Raw recording exhibiting low signal-to-noise ratio and reduced spatial contrast. Right, NeuroPilot-restored image reconstructed from the same acquisition, showing enhanced contrast, improved signal-to-background ratio, and recovery of fine vascular and neuronal structures across the wide field of view. Insets display magnified views of the boxed regions. Upper corners show representative intensity profiles along the indicated dashed lines. Bottom: Kymographs illustrating improved temporal signal consistency. Scale bars: 1 mm (main), 200 μm (insets). Time scale: 10 s.



Supplementary Figure 5

Restoration of spatiotemporal fidelity in multiplexed imaging channels and diverse biological samples by NeuroPilot.

a, Representative dual-channel two-photon imaging of the mouse vibrissal primary somatosensory cortex (vS1) **TODO cite** acquired at 7 Hz.

Raw (left) and NeuroPilot-restored (right) images display GCaMP6s (green, calcium activity) and mCherry (magenta, nuclei marker) signals. Boxed insets (1 and 2) reveal improved resolution of fine structural features after denoising. Kymographs extracted along the indicated dashed lines (vertical panels) demonstrate preservation of temporal dynamics with reduced background fluctuations. Scale bars: 50 μm (main), 10 μm (insets), 1 s (kymograph).

b, Spatial footprints of segmented neuronal ROIs across the field of view. Each pseudocolor represents an individual neuron.

c, Representative calcium fluorescence dynamics from selected neurons. Traces correspond to the numbered cells in **b**. Overlay of raw (gray) and NeuroPilot-processed (blue) traces illustrates the attenuation of background noise. Time scale: 50 s.

d, Quantitative assessment of signal fidelity. Violin plots displaying the Pearson correlation coefficients between the extracted traces.

e, Local spatial correlation analysis of signal restoration. Comparative local Pearson correlation maps for raw (lower-left triangle) and NeuroPilot-processed (upper-right triangle) data, for both GCaMP6s and mCherry channels.

f, NeuroPilot performance on spinal cord vascular imaging. Comparison between raw two-photon recordings (left) and the corresponding NeuroPilot-restored results (right). Top panels show representative fields of view with magnified insets, highlighting improved continuity and contrast of vascular structures. Middle panels display kymographs extracted along the indicated dashed lines, showing reduced motion artifacts and improved temporal consistency. Scale bars: 25 μm (main), 10 μm (insets) and 50 μm (MIP). Time scale bar: 10 s.

g, Representative comparison of raw and NeuroPilot-processed mitochondrial calcium imaging. Insets show magnified views of the dashed regions. Intensity profiles (top right) are extracted along the indicated lines.

h, Local correlation maps comparing raw (lower-right) and NeuroPilot-processed (upper-left) data.

i, Mitochondrial segmentation and skeletonization using the MINA pipeline. Distinct colors denote endpoints, junctions, branches, and outlines, revealing improved structural continuity in NeuroPilot outputs.

162 **j**, Temporal intensity profiles extracted along the indicated lines, demonstrating reduced
163 signal fluctuation and improved temporal stability after NeuroPilot processing. Red
164 arrowheads indicate time points with pronounced motion artifacts in the raw recordings.
165 **k**, Quantitative analysis of mitochondrial morphology. NeuroPilot significantly
166 increases mitochondrial footprint, network branching, and branch length compared to
167 raw data.

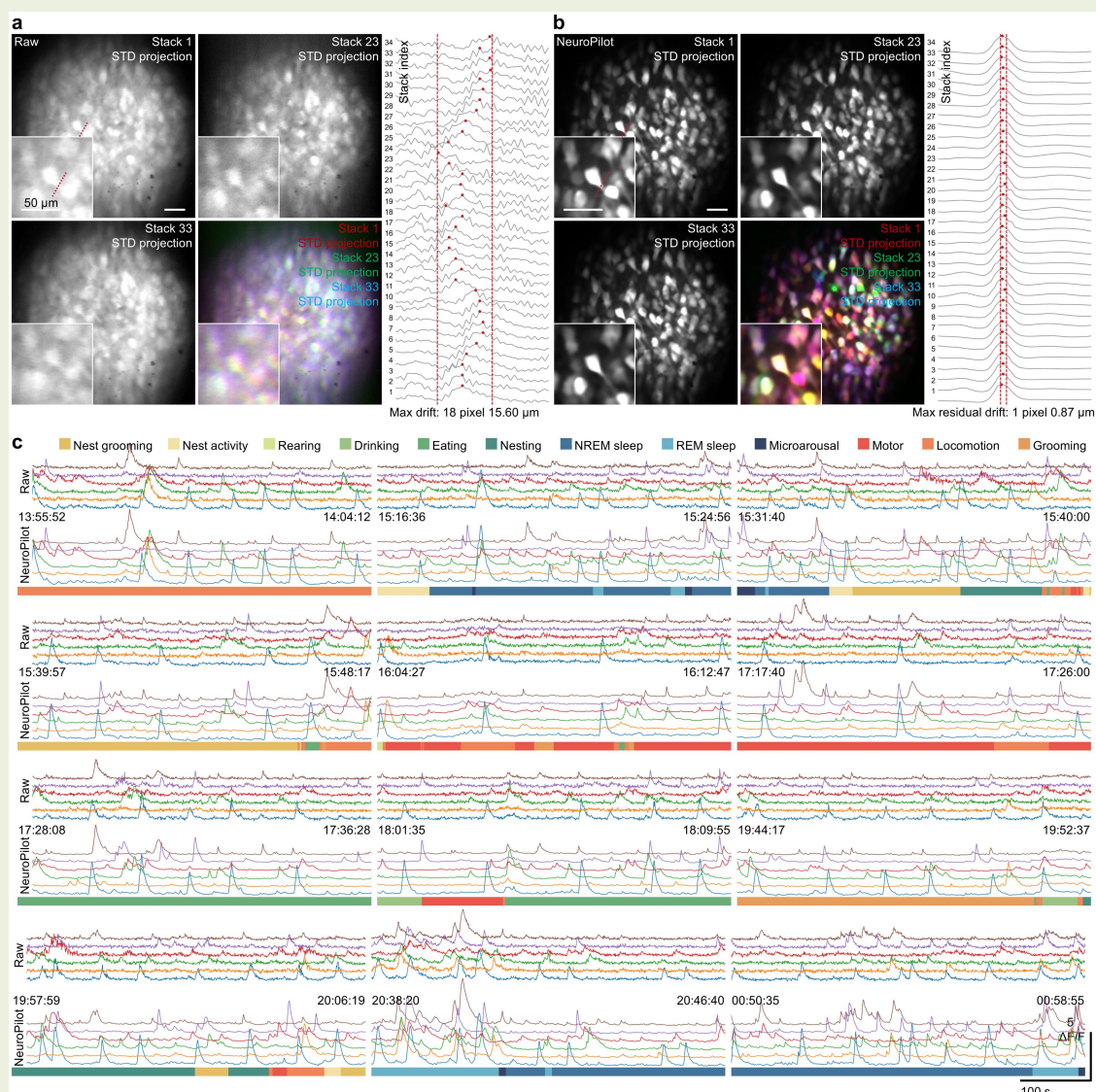


Supplementary Figure 6

Long-term 24-hour tracking of individual mPFC neurons and calcium dynamics via NeuroPilot.

a, Representative field of view (FOV) from a 24-hour head-mounted two-photon recording in the mouse mPFC. Insets show magnified views of six individual neurons with their corresponding segmented regions of interest (ROIs).

b, Time-resolved calcium traces for the six neurons identified in (a) over a continuous 24-hour period. Traces are color-coded to match their respective ROIs. The sequential panels represent concatenated recording segments with absolute time stamps indicated. Simultaneous behavioral states are displayed as a color-coded ethogram below the calcium traces (see legend for state definitions). Scale bars: 50 μm in **a**; 5 $\Delta F/F$ and 200 s in **b**.



Supplementary Figure 7

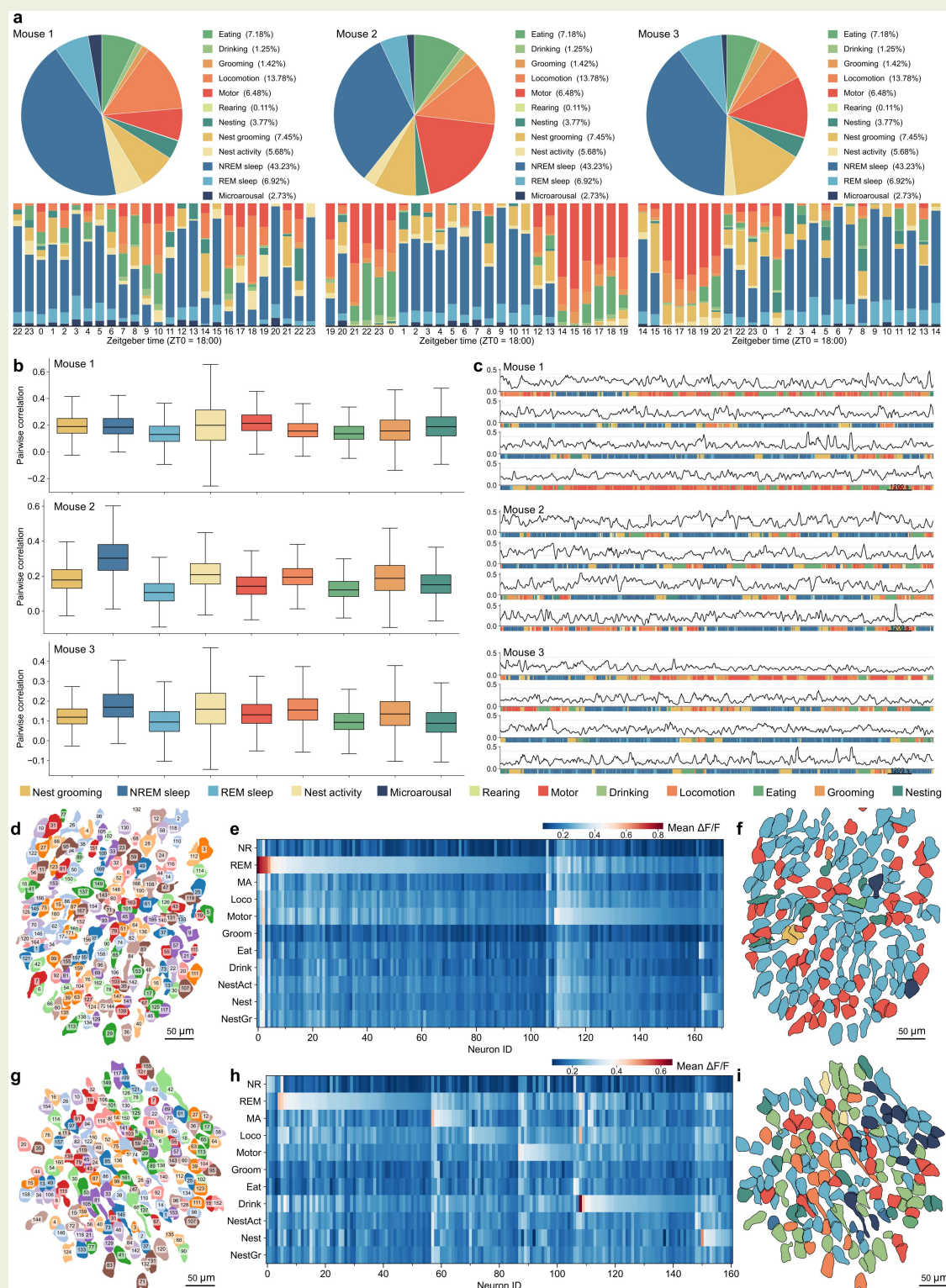
Long-term stabilization and preservation of spatiotemporal neural activity across behavioral states enabled by NeuroPilot.

a, STD projections computed from selected imaging stacks across a 24-hour recording before correction. Representative stacks (1, 23, and 33) and a composite RGB overlay reveal severe motion artifacts. Line profiles extracted across the indicated somatic features (dashed lines, insets) illustrate substantial stack-to-stack spatial drift (up to 15.60 μ m).

b, Corresponding STD projections of the exact same field of view following NeuroPilot correction. Individual stacks and the converged RGB composite demonstrate restored structural fidelity. Extracted line profiles confirm robust, sub-pixel spatial alignment

(within 0.87 μm) over the entire 24-hour period. Red dashed lines indicate reference positions. Insets provide magnified views of the highlighted regions.

c, Representative $\Delta F/F$ traces from a subset of mPFC neurons, comparing raw recordings (top) with NeuroPilot-processed outputs (bottom) across extended sessions. Time-lapse segments are temporally aligned with color-coded behavioral state annotations, illustrating diverse neural activity patterns during transitions between states. Scale bars: 5 $\Delta F/F$, 100 s.



Supplementary Figure 8

Behavior-specific activity patterns and population correlation structure of mPFC neurons across distinct behavioral states.

a, Behavioral composition for three representative animals (mouse 1-3). Top, pie charts showing the overall fraction of time spent in each behavioral state. Bottom, stacked bar plots illustrating the temporal distribution of behavioral states across Zeitgeber time (ZT; ZT0 = 18:00). Each bar represents one-hour bins, with colors indicating distinct behavioral states.

b, Box plots show the distribution of pairwise Pearson correlation coefficients computed using a 60-s sliding window across behavioral states (colors denote behaviors; median and interquartile range indicated).

c, Population-averaged pairwise correlation dynamics. Correlations were computed using a 60-s sliding window and averaged across neuron pairs. Traces are shown across consecutive recording epochs with aligned behavioral annotations. Scale bars, 1200 s.

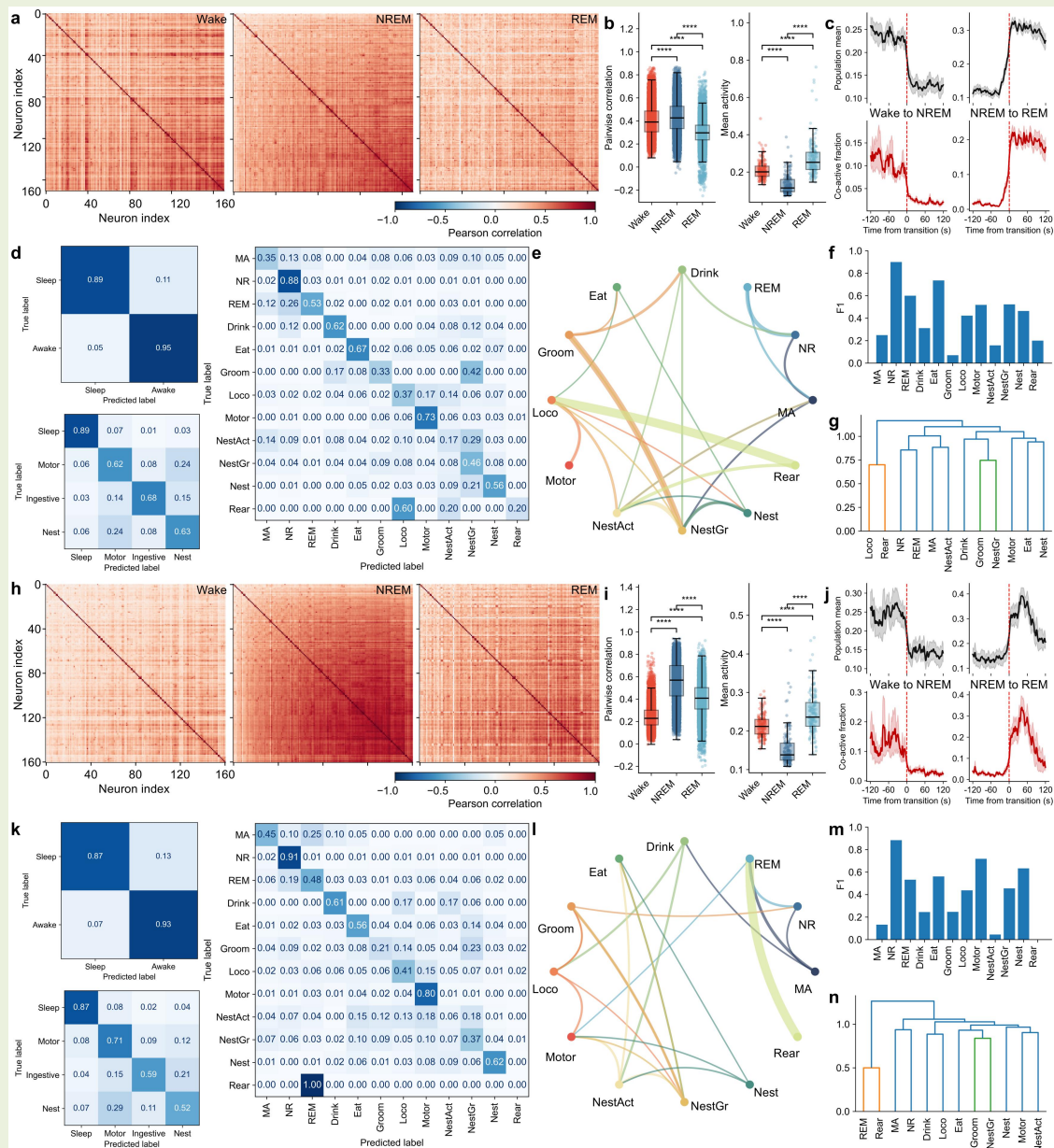
d-f, Representative mPFC population activity from an individual mouse (mouse 2; n = 170 neurons).

d, Spatial map of segmented neuronal somata within the imaging field.

e, Mean calcium responses ($\Delta F/F$) of individual neurons across behavioral states, sorted by their dominant activity preference.

f, Spatial distribution of neurons color-coded by the behavioral state exhibiting the highest mean $\Delta F/F$ response.

g-i, Same analysis for a second mouse (mouse 3; n = 160 neurons).



Supplementary Figure 9

State-dependent correlation structure and hierarchical decoding of mPFC population activity with transition dynamics.

a, Pairwise Pearson correlation matrices of mPFC population activity during wake, NREM, and REM states. Neurons are ordered identically across conditions.

b, Summary statistics of pairwise correlations and mean neuronal activity across behavioral states. Box plots indicate median and interquartile range.

c, Transition-aligned population dynamics of mPFC activity. Left, transitions from wake to NREM; right, transitions from NREM to REM. Top, mean population activity

averaged across neurons; bottom, co-active fraction defined as the proportion of neurons exceeding an activity threshold at each time point. Traces represent mean $\pm$ s.e.m. across transitions. Red dashed lines indicate transition onset ($t = 0$).

d, Decoding performance across hierarchical behavioral resolutions. Top left, binary classification of sleep versus wake states; bottom left, coarse behavioral categorization; right, fine-grained decoding across 12 behavioral states. Values indicate normalized prediction accuracy.

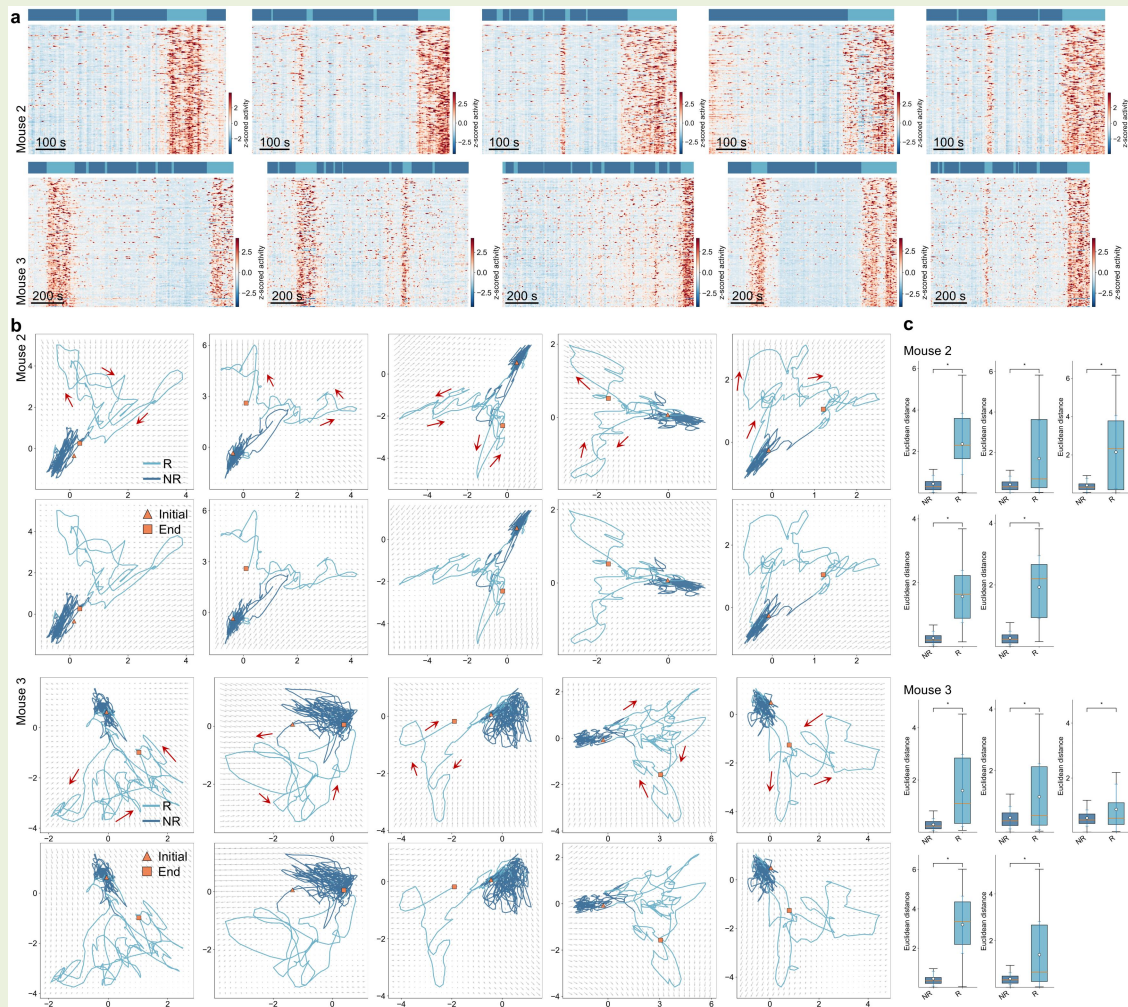
e, Behavioral state transition network inferred from temporal sequences, where nodes represent behavioral states and edge thickness denotes transition probability.

f, Decoding performance across behavioral states quantified by F1 scores for each category.

g, Hierarchical clustering dendrogram illustrating the similarity structure of behavioral representations in the mPFC population activity space.

a-g, Analyses from mouse 2.

h-n, identical analyses applied to mouse 3.



Supplementary Figure 10

Latent mPFC population dynamics and attractor-like organization during NREM-to-REM sleep transitions.

a, Heatmaps showing representative z-scored mPFC population activity aligned to spontaneous NREM-to-REM transitions. Each row corresponds to a single neuron, and columns represent time aligned to the transition event. The top color bar indicates annotated sleep states (dark blue, NREM; light blue, REM). Examples from multiple animals (mouse 2 and mouse 3) are shown.

b, Latent mPFC population trajectories corresponding to the events shown in **a**, inferred using recurrent switching linear dynamical systems (rSLDS). Trajectories are color-coded by behavioral state (dark blue, NREM; light blue, REM), with initial (orange triangles) and terminal (orange squares) points indicated. The same trajectory is overlaid on two distinct vector fields (grey arrows) corresponding to separate dynamical

257 modes. Red arrows indicate representative flow directions. Examples from mouse 2 and
258 mouse 3 are shown.

259 **c**, Quantification of the Euclidean distance between instantaneous latent states and the
260 corresponding attractor states. Distances are compared between NREM (NR) and REM
261 (R), showing increased dispersion relative to the attractor during REM. Box plots
262 indicate median and interquartile range. Data from mouse 2 and mouse 3.